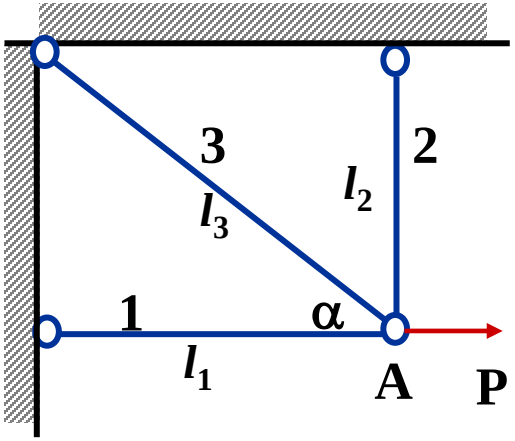
第二十八讲（2）

思考题讨论

作业：14-7(b),9(a),14,11,
15-15

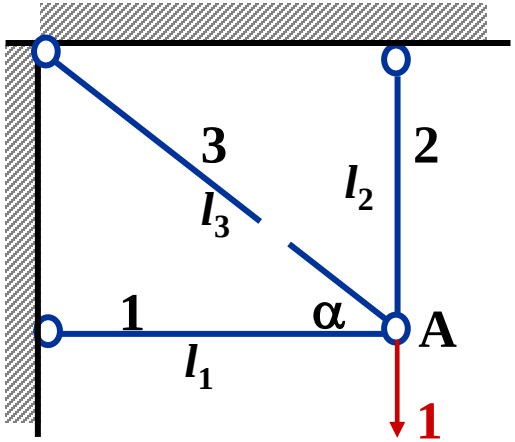
思考题

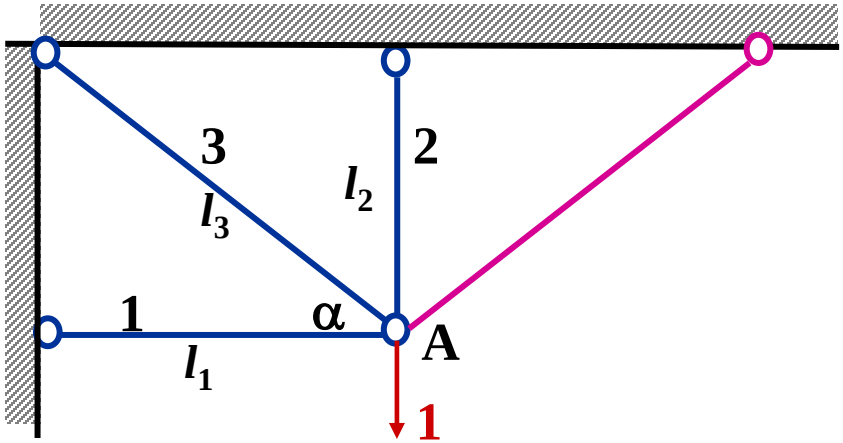
图示桁架，各杆*EA*相同，求*A*点的铅垂位移。



在求出各杆轴力后，求A点的铅垂位移：

选取单位载荷状态：





单位力加在基本系统上，为什么？

该单位载荷状态是否可用？

关于变形体虚功原理

满足平衡方程、静力边界条件

满足变形协调条件、位移边界条件

外力 $q(x)$

内力 $M(x)$

静力许可场

虚位移 $w^*(x)$

虚变形 $\mathrm{d}\theta^*$

运动许可场

彼此独立

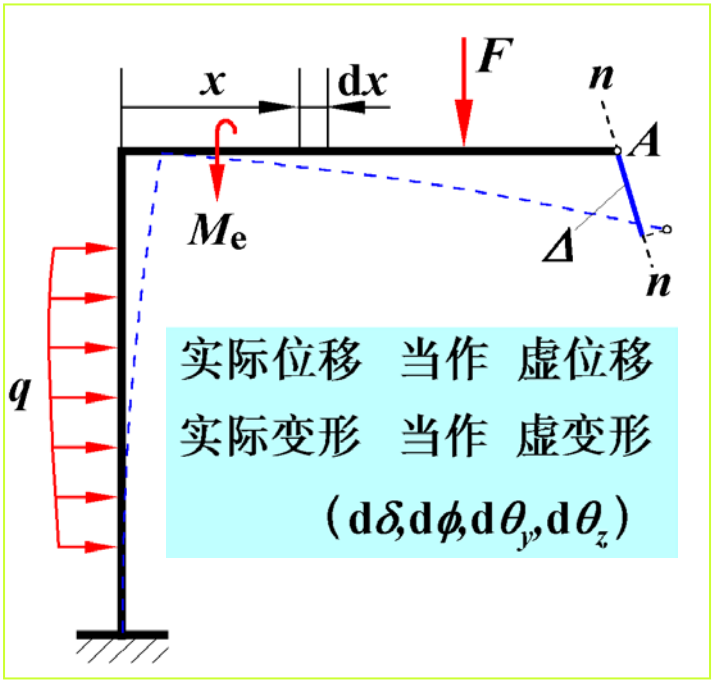
外力虚功 W_e

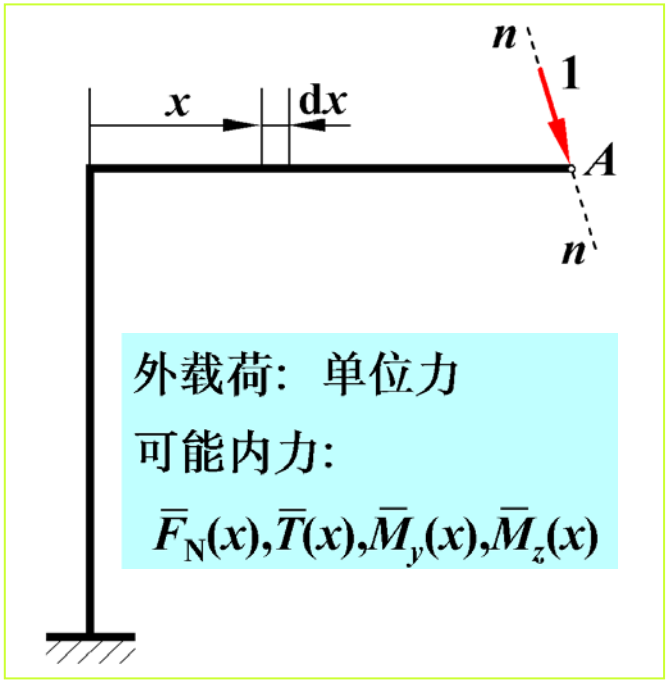
内力虚功 W_i

适用于线弹性、非线弹性、非弹性体

单位载荷法

结构的载荷状态





构造单位（载荷）状态

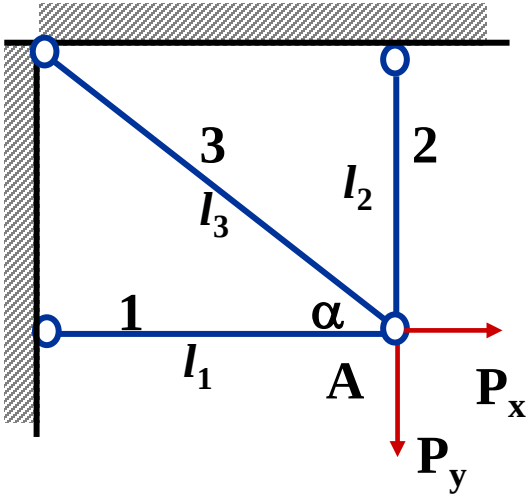
Δ --- $d\delta, d\phi, d\theta_y, d\theta_z$ → 运动许可场

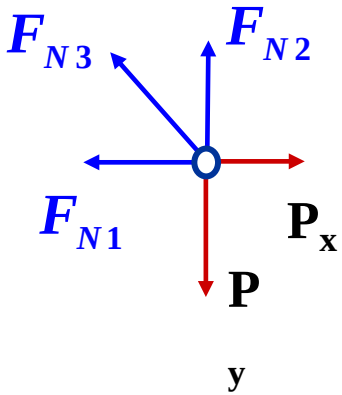
1 --- $\bar{F}_N, \bar{T}, \bar{M}_y, \bar{M}_z$ → 静力许可场

$$1\cdot\Delta = \int_l \left[\bar{F}_N(x)d\delta + \bar{T}(x)d\phi + \bar{M}_y(x)d\theta_y + \bar{M}_z(x)d\theta_z\right]$$

变形体虚功原理

可能内力





$$\begin{cases} F_{N1} + F_{N3}\cos\alpha = P_x \\ F_{N2} + F_{N3}\sin\alpha = P_y \end{cases}$$

平衡方程有无数内力解答；

任一组内力解答均为可能内力

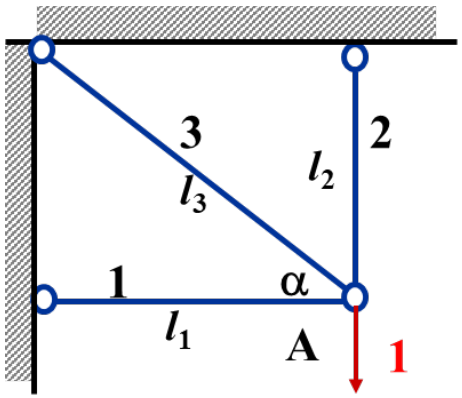
静不定结构有无数多可能内力

思考题

单位力加在原静不定结构上

可能内力

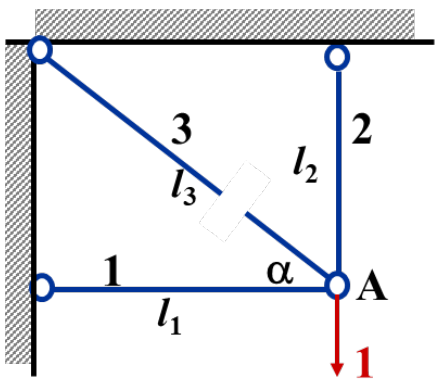
$$\begin{cases} F_{N1} + F_{N3}\cos\alpha = 0 \\ F_{N2} + F_{N3}\sin\alpha = 1 \end{cases}$$



单位力加在基本系统上

$F_{N2} = 1,$

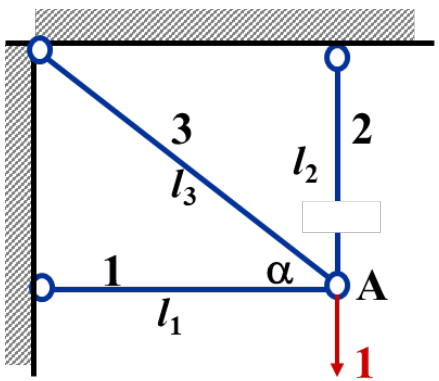
$F_{N1} = F_{N3} = 0$



$F_{N2} = 0,$

$F_{N3} = \dfrac{1}{\sin\alpha}$

$F_{N1} = -\tan\alpha$



节点平衡方程满足，内力是可能内力，

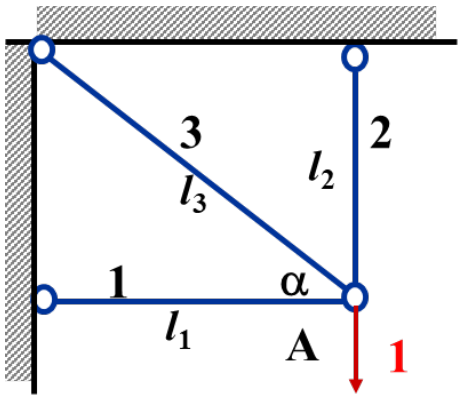
单位载荷状态可用

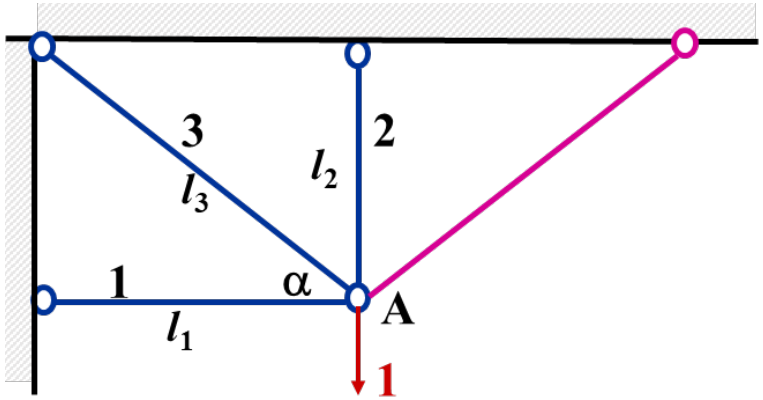
思考题

单位力加在原静不定结构上

可能内力

$$\begin{cases} F_{N1} + F_{N3}\cos\alpha = 0 \\ F_{N2} + F_{N3}\sin\alpha = 1 \end{cases}$$





$$\begin{cases} F_{N1} + F_{N3}\cos\alpha - F_{N4}\cos\alpha = 0 \\ F_{N2} + F_{N3}\sin\alpha + F_{N4}\sin\alpha = 1 \end{cases}$$

只要4杆轴力不为零，1、2、3杆轴力不满足原结构节点平衡方程，内力不是可能内力，单位载荷状态不可用。

本讲：思考题讨论

单位载荷状态构造了静力许可场

下讲：